

1 Baseline

Modèle d'optimisation existant en l'état.

1.1 Objective:

N units i in which investors decide in t to invest d_{it} and to produce x_{it} over $T - t_0$ periods
 c_i extraction cost, C_i full development cost, θ_i carbon intensity
 X_t demand in t , p_t^{ref} and X_t^{ref} reference price and quantity for demand curve in t , ϵ_t elasticity in t (negative, magnitude increasing over time from 0.1 to 0.6).
 ρ discount rate

$$\begin{aligned}
 Min_{x_{it}, d_{it}} & \sum_{i=1}^N \sum_{t=t_0}^T (c_i + \tau \theta_i e^{\rho t}) x_{it} e^{-\rho t} && \text{Production} \\
 & + \sum_{i=1}^N \sum_{t=t_0}^T C_i d_{it} e^{-\rho t} && \text{Investment} \\
 & - \sum_{t=t_0}^T \frac{\epsilon_t}{1 + \epsilon_t} X_t^{\frac{1+\epsilon_t}{\epsilon_t}} p_t^{ref} (X_t^{ref})^{-\frac{1}{\epsilon_t}} e^{-\rho t} && \text{Consumer surplus}
 \end{aligned}$$

1.2 Constraints:

1. Production capacity (Decline) + Reserves:

$$\forall_{i,t} \sum_{u=t_0}^{t-1} x_{iu} + \frac{x_{it}}{DR_i} \leq R_i$$

with DR_i Decline rate of i and R_i available reserves in 1992.

2. **Without depreciation:** Development + Production capacity (plateau, proportional to development level):

$$\forall_{i,t} d_{i\underline{t}} + \sum_{u=t_0}^{t-LAD_i} d_{iu} - \frac{x_{it}}{\bar{x}_i} \geq 0$$

with $d_{i\underline{t}} = \sum_{u=-1}^{t_0-1} d_{iu}$ (development level in t_0), LAD_i lag after development (lag between investment and production capacity increase; can be i specific or not), and $\bar{x}_i$ plateau production.

With depreciation:

$$\forall_{i,t} d_{i\underline{t}} + \sum_{u=t_0}^{t-LAD_i} (1 - \delta_i)^{t-u} d_{iu} - \frac{x_{it}}{\bar{x}_i} \geq 0$$

with δ_i depreciation rate of capital of asset i.

3. Maximum development level (constrained to 1):

$$\forall_i \ 0 \leq d_{i\underline{t}} + \sum_{t=t_0}^T d_{it} \leq 1$$

4. Demand:

$$\forall_t \sum_{i=1}^N x_{it} \geq X_t$$

5. Discovery year:

$$\forall t < t_i^{disc}, d_{it} = 0$$

with t_i^{disc} discovery year of i.